

1. Administrative & Study Identification

Full Study Title: A Study to Evaluate Treatment Patterns, Outcomes, and Healthcare Resource Utilization in Patients With Lower-Risk Myelodysplastic Syndromes Treated With Luspatercept
Official Regulatory Title: Real-world Utilization Patterns, Clinical Outcomes, and HCRU in Lower-risk MDS Patients Treated With Luspatercept: A Multinational Medical Record Review Study
Short Title/Acronym: Luspatercept Real-World LR-MDS Study
Protocol Number: CA056-1043
NCT Number: NCT07075406
Posting ID: 1046382081401408161
Sponsor/Company Name: Bristol Myers Squibb
Investigational Product: Luspatercept (Trade Name: Reblozyl) |
ATC Code: B03XA06
Phase of Development: Not Applicable
Trial Status: COMPLETED
Medical Monitor: Bristol Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To understand the treatment use of luspatercept in adults diagnosed with lower-risk myelodysplastic syndromes (LR-MDS), evaluating real-world treatment utilization patterns, clinical outcomes, and healthcare resource utilization (HCRU).
Secondary Objectives: To characterize patient demographics, evaluate real-world dosing and duration, and assess clinical outcomes such as transfusion independence and hematologic improvement in a real-world setting.

2.2 Background & Rationale To understand the treatment use and real-world effectiveness of luspatercept in adults diagnosed with lower-risk myelodysplastic syndromes (LR-MDS). By collecting and reviewing medical histories, this study provides valuable insights into how luspatercept fits into current treatment strategies, ultimately aiming to improve health outcomes and quality of life for LR-MDS patients in routine clinical practice.

3. Study Design & Methodology

3.1 Design Framework Study Type: OBSERVATIONAL (Multinational Medical Record Review Study)
Control Type: Single-arm cohort (Observational)
Blinding: Open-label / N/A (Retrospective Review)
Randomization: N/A

3.2 Planned Enrollment & Duration Targeted Enrollment: 250 patients

Number of Sites: Multinational (Including US, Germany, Spain, France, Canada)

Estimated Study Start: 01-Nov-2024

Estimated Primary Completion: 01-Aug-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Confirmed diagnosis of primary myelodysplastic syndromes (MDS) with lower-risk status as measured by the International Prognostic Scoring System (IPSS) or the Revised International Prognostic Scoring System (IPSS-R) at the time of diagnosis: * IPSS risk level: low, intermediate-1 (level-1 risk) * IPSS-R risk level: very low, low, intermediate 2. Initiated luspatercept for treatment of Lower-Risk (LR)-MDS after the initial availability in each country of interest: * US: after April 2020 * Germany: after June 2020 * Spain: after June 2020 * France: after June 202 * Canada: after February 2021 3. The participant has a potential follow-up of at least 6 months from the index date (except death). 4. The participant is aged 18 years or older at the index date. 5. The participant has a complete medical record or history for at least 12 months before the index date (or up to the date of initial LR-MDS diagnosis if duration between initial diagnosis and index date is less than 12 months).

4.2 Exclusion Criteria 1. Received luspatercept as part of a clinical trial. 2. The participant has evidence of other malignant neoplasms prior to diagnosis of MDS, except disease free for ≥ 5 years at time of MDS diagnosis, basal or squamous cell carcinoma of the skin, carcinoma in situ of the cervix, carcinoma in situ of the breast, or incidental histologic finding of prostate cancer (stage T1a or T1b). 3. The participant has a history of Acute Myeloid Leukemia prior to MDS diagnosis. 4. The participant has participated in clinical trials for specific treatments related to treatment of MDS.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Routine clinical practice prescribing (Real-world dosing of Luspatercept).

Route of Administration: Subcutaneous (as per standard of care label).

Duration of Treatment: Variable based on real-world continuous treatment durations captured in medical records.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for MDS and co-morbidities as prescribed by the treating physician in routine care.

Prohibited: Concurrent participation in interventional clinical trials for MDS.

6. Study Timeline & Visit Schedule

(Note: As a retrospective medical record review, standard clinical visit windows do not apply. The following reflects the abstraction timeline)

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening / Identification	Retrospective Identify eligible patient charts based on diagnosis and luspatercept initiation.
2	Baseline Data Abstraction	Retrospective Collect demographics, IPSS/IPSS-R scores, and prior MDS treatments.
3	Treatment Follow-up Review	Retrospective Abstract real-world dosing, dose modifications, HCRU, and response data.
4	End of Study / Exit	Retrospective Record reasons for discontinuation, overall survival status, or last known follow-up date.

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Participant baseline demographics 2. Participant baseline clinical characteristics 3. Participant treatment history

7.2 Secondary Outcomes 1. Number of participants receiving myelodysplastic syndromes (MDS) -directed treatment modalities 2. Systemic therapy agent or combination of agents received by participants 3. Treating clinicians' rationale for prescribing treatment(s)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Retrospective abstraction of AEs documented by the treating physician in the patient's medical records.

Serious Adverse Events (SAEs): Captured historically through HCRU metrics (e.g., hospitalizations, emergency room visits) and chart notes.

Data Monitoring Committee (DMC): N/A (Observational, non-interventional study).

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All eligible LR-MDS patients whose medical records are successfully abstracted (\$N=250\$).

Sample Size Justification: Target sample of 250 patients is deemed sufficient to provide descriptive statistical power for real-world usage patterns and HCRU trends.

Handling of Missing Data: Standard handling for observational real-world evidence (e.g., missing data will not be imputed unless otherwise specified; denominators will reflect the number of patients with available data).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Pharmacoepidemiology Practices (GPP) and local data privacy regulations (e.g., HIPAA, GDPR).

Monitoring Plan: Routine quality control checks of abstracted data against source medical records to ensure accuracy.

Audit Trail: Secure database abstraction environments utilizing eCRF with comprehensive audit trails.

11. Study Identifiers & Governance

Primary ID: CA056-1043 **Registry Number:** NCT07075406 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate Treatment Patterns, Outcomes, and Healthcare Resource Utilization in Patients With Lower-Risk Myelodysplastic Syndromes Treated With Luspatercept **Official Regulatory Title:** Real-world Utilization Patterns, Clinical Outcomes, and HCRU in Lower-risk MDS Patients Treated With Luspatercept: A Multinational Medical Record Review Study

12. Clinical Framework (MDS Specific)

Disease Area: Myelodysplastic Syndromes **Preferred UMLS Name:** Lower-Risk Myelodysplastic Syndromes **MeSH Code:** D009190 **SNOMED ID:** 109995007 **Diagnosis Threshold:** IPSS risk level (low, intermediate-1) or IPSS-R risk level (very low, low, intermediate) **Medical Methodology:** Retrospective Chart Review / Real-World Evidence **Optimization Target:** Anemia management / Transfusion independence via Luspatercept therapy

13. Study Procedures & Methodology

Management Pathway: Routine Clinical Practice for LR-MDS **Education Protocol (HCP):** N/A (Observational Study) **Education Protocol (Patient):** N/A (Observational Study) **Quality Audit Mechanism:** Automated data validation checks during record abstraction and targeted source data verification

14. Observation & Follow-Up Schedule

Total Duration: Retrospective look-back period (from availability date, e.g., April 2020 in the US, to abstraction date) **Onsite**

Visit Cadence: N/A (Retrospective data abstraction) **Remote**

Monitoring Frequency: N/A (Retrospective data abstraction)

Checklist Review Interval: Continuous retrospective review over the patient's documented treatment lifespan

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					